



**RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF TECHNOLOGY**

**Bachelor of Information and Communication Technology
First Year - Semester I Examination – February/March 2023**

ICT 1305 – PROGRAM DESIGNING AND PROGRAMMING

Time: Three (03) hours

Answer **all** Questions in Part A and B

Part A

Write down the letter (a, b, c, d) of most appropriate answer with the question number in the answer book.

E.g.: 1. a

1. Correct way of commenting a single line in C programming is
 - a) `/*printf("Hello C..");`
 - b) `//printf("Hello C..");`
 - c) `/*printf("Hello C..");*/.`
 - d) `/printf("Hello C..");/`
2. Operator % in C programming Language is called
 - a) Percentage Operator
 - b) Quotient Operator
 - c) Modulus Operator
 - d) Division Operator

3. Choose the correct statement.

- a) `int my_age = 10;`
- b) `int lmy,age = 10;`
- c) `int my age = 10;`
- d) All of the mentioned

4. How many iterations in the while of the following C program?

```
#include <stdio.h>

int main()
{
    int i = 0;
    while (i < 3)
        i++;
    printf("In while loop\n")
}
```

- a) 2
- b) 3
- c) 4.
- d) 1

5. What are the elements present in the array of the following C code?

```
int array[5] = {5};
```

- a) 5, 5, 5, 5, 5
- b) 5, 0, 0, 0, 0
- c) 5, (garbage), (garbage), (garbage), (garbage)
- d) (garbage), (garbage), (garbage), (garbage), 5

6. What will be the output of the following C code?

```
#include <stdio.h>
int main()
{
    int i = 5;
    i = i / 3;
    printf("%d\n", i);
    return 0;
}
```

- a) 0
 - b) 1
 - c) 3
 - d) Compile time error
7. Which of the following cannot be a structure member?
- a) Another structure
 - b) Function
 - c) Array
 - d) None of the mentioned
8. Algorithms can not be represented as
- a) pseudo codes
 - b) Syntax
 - c) programs
 - d) Flowcharts
9. The function **scanf** is used to
- a) Take logical decisions
 - b) Input a set of values
 - c) Print a set of values
 - d) Do mathematical manipulations

10. If the function returns no value, then it is called

- a) Data type function
- b) Calling function
- c) Main function
- d) Void function

11. What will be the final value of x in the following C code?

```
#include <stdio.h>
void main()
{
    int x = 5 * 9 / 3 + 9;
}
```

- a) 3.75
- b) Depends on compiler
- c) 24
- d) 3

12. Which of the following is a pictorial representation of an algorithm.

- a) Pseudo code
- b) Program
- c) Flowchart
- d) Algorithm

13. What is the symbol used to represent output in a flowchart .

- a) Square
- b) Circle
- c) Parallelogram
- d) Diamond

14. Which of the following can be replaced by if statement.

- a) Switch
- b) While
- c) Continue
- d) For

15. The mechanism used to convey information to the function is the

- a) Argument
- b) Commands
- c) Loops
- d) Statements

16. What will be the output of the following C code?

```
#include <stdio.h>
int main()
{
    int y = 10000;
    int y = 34;
    printf("Hello World! %d\n", y);
    return 0;
}
```

- a) Compile time error
- b) Hello World! 34
- c) Hello World! 1000
- d) Hello World! followed by a junk value

17. What is the value of variable *a* after executing following statement.

```
int a = 10 + 4.867;
```

- a) 10
- b) 14.867
- c) 14
- d) compiler error.

18. Choose the correct statement about Functions in C Language.

- a) A Function is a group of C statements which can be reused any number of times.
- b) Every Function has a return type.
- c) Every Function may or may not return a value.
- d) All of the above

19. A function which calls itself is called a ____ function.

- a) Self
- b) Auto
- c) Recursive
- d) Static

20. In C, an array index starts with

- a) -1
- b) 0
- c) 1
- d) 2

[Total 20 marks]

Part B

1.

a) What is an algorithm? [03 marks]

b) Compare pseudo codes vs flowcharts by highlighting their advantages and disadvantages.

[05 marks]

c) What is "Desk checking" in programming?

[02 marks]

d) Draw a flowchart for the given algorithm .

1. Start the program

2. Initialize variables $i=0, \max=0$ 3. Read n 4. If $n > \max$, then $\max = n$ 5. Increment i by 16. If $i < 50$, go to step 3, otherwise go to step 77. Print $\max$

8. End

[10 marks]

[Total 20 marks]

2.

a) Complete the following function to calculate sum of integers from 0 to n .`int summation (_____) {``int sum=0;``for(int i=0; _____; i++){``_____``_____;``}`

[05 marks]

b) Write the function given in part (a) as a recursive function

.[06 marks]

- c) Write a C program to print the following star pattern.

```
*****
```

```
****
```

```
***
```

```
**
```

```
*
```

[10 marks]

- d) What is a structure and a union in C programming language?

[04 marks]

[Total 25 marks]

3.

- a) Complete the following code with struct data structure .

```
struct Book {
    char title[15];
    int id;
    char category;
};

int main() {
    _____;
    strcpy(_____, "ABC");
    s1.id = 13;
    s1.category = 'B';
    printf(_____, s1.title);
    printf(_____, s1.id);
    printf("category: %c\n", _____);
    return 0;
}
```

[05 marks]

- b) What will be the output of the below code.

```
#include <stdio.h>

void main()
{
    int i,n,sum=0;
    scanf("%d",&n);
    printf("\nThe numbers are :");
    for(i=1;i<=n;i++)
    {
        printf("%d ",2*i-1);
        sum=sum+2*i-1;
    }
    printf("\nThe Sum upto %d terms : %d \n",n,sum);
}
```

[03 marks]

- c) Write a C program that accepts an integer from the user and prints whether the integer is an even number or an odd number.

[07 marks]

[Total 15 marks]

4.

- a) Write an algorithm to find the multiplication of 10 numbers entered by the user.

[05 marks]

- b) Draw a flowchart which reads 10 numbers from the user and calculate the sum of the even numbers entered by the user.

[07 marks]

- c) Draw a flowchart to read 10 exam scores from the user and print "Pass" or "Fail" accordingly, if the pass mark is 60. Your flowchart should count and print the number of Passes and, average of the pass marks

[08 marks]

[Total 20 marks]

--End--